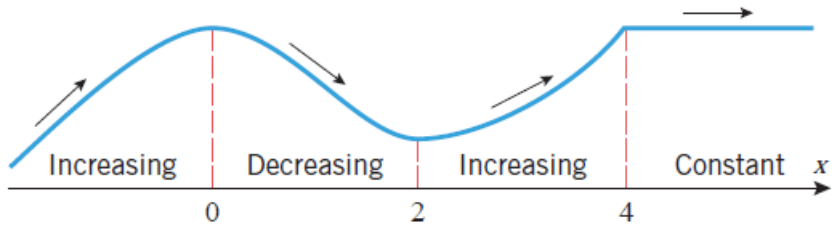
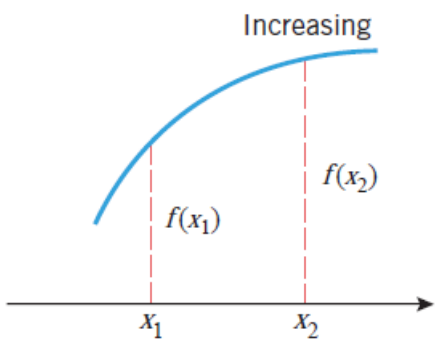
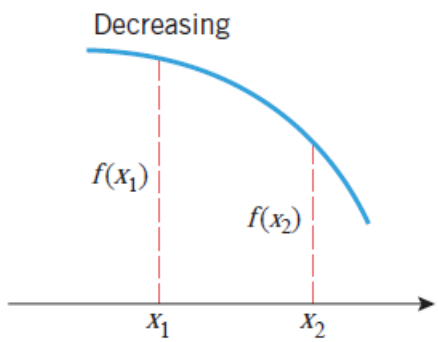
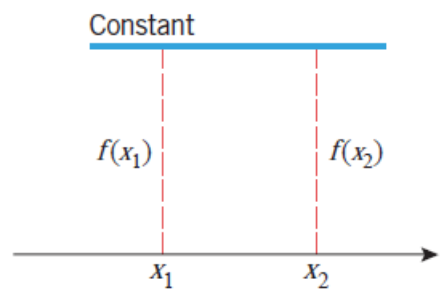
Maxima and Minima









Each tangent line has a +ve slope. Each tangent line has a -ve slope. Each tangent line has zero slope.

(Consequence of the Mean Value Theorem (MVT))

Theorem: Let f be a function that is continuous on a closed interval $[a, b]$ and differentiable on the open interval (a, b). Then,

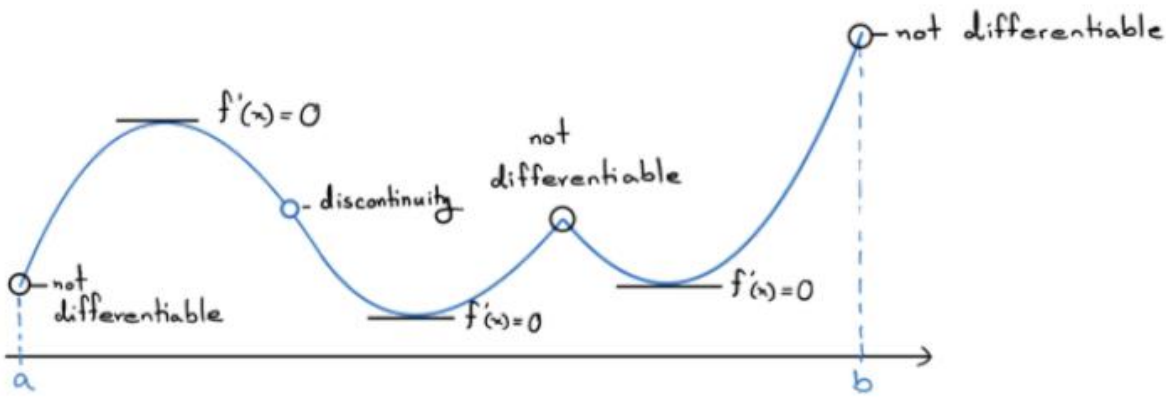
1. If $f'(x) > 0$ for every value of x in (a, b), then f is increasing on $[a, b]$.
2. If $f'(x) < 0$ for every value of x in (a, b), then f is decreasing on $[a, b]$.
3. If $f'(x) = 0$ for every value of x in (a, b), then f is constant on $[a, b]$.

<u>Critical Point and Stationary Point:</u>

<u>Critical Points:</u> Critical points are points where the derivative is either zero or undefined (not differentiable). Meaning $f'(x_0) = 0$ or $f(x)$ is not differentiable at $x = x_0$. So, turning points, end points of the interval on which $f(x)$ is defined and points of discontinuity are all critical points.

<u>Stationary Points:</u> A stationary point is a point where the derivative is zero and only zero. Meaning,

$f'(x_0) = 0$ at $x = x_0$.



All the labelled points are critical points. Points with $f'(x) = 0$ are also stationary points.

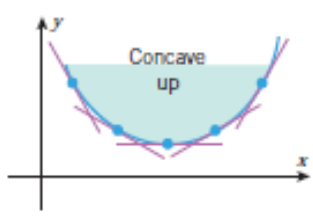
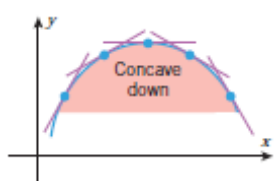
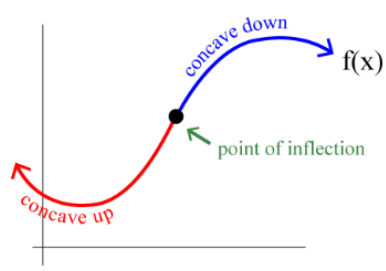
[**Note:** When finding critical points for polynomials, the only critical points occur when $f'(x) = 0$, as polynomials are differentiable everywhere.

Note: If $f(x)$ is not defined on a fixed interval, then there are no endpoints.]

<u>Concavity:</u> (Represents the direction in which a graph curves)

<u>Theorem:</u> Let f be twice differentiable on an open interval (a, b). Then,

i. If $f''(x) > 0$, for every value of x in the open interval (a, b), then f is concave up (the graph curves upwards) on that interval (a, b).

ii. If $f''(x) < 0$, for every value of x in the open interval (a, b), then f is concave down (the graph curves downwards) on that interval (a, b).

iii. If $f''(x) = 0$, then f has an inflection point.